

Physical Chemistry (I)

Lecture 04

Internal Energy

Real Gas



In the Last Class

- Maxwell-Boltzmann distribution: $E \rightarrow \vec{v} \rightarrow v$
 - Characteristic speeds
- Collisions
 - Molecule-molecule collisions: chemical kinetics
 - Molecule-wall collisions: pressure

Last class

Maxwell-Boltzmann Distribution

- Velocity distribution:

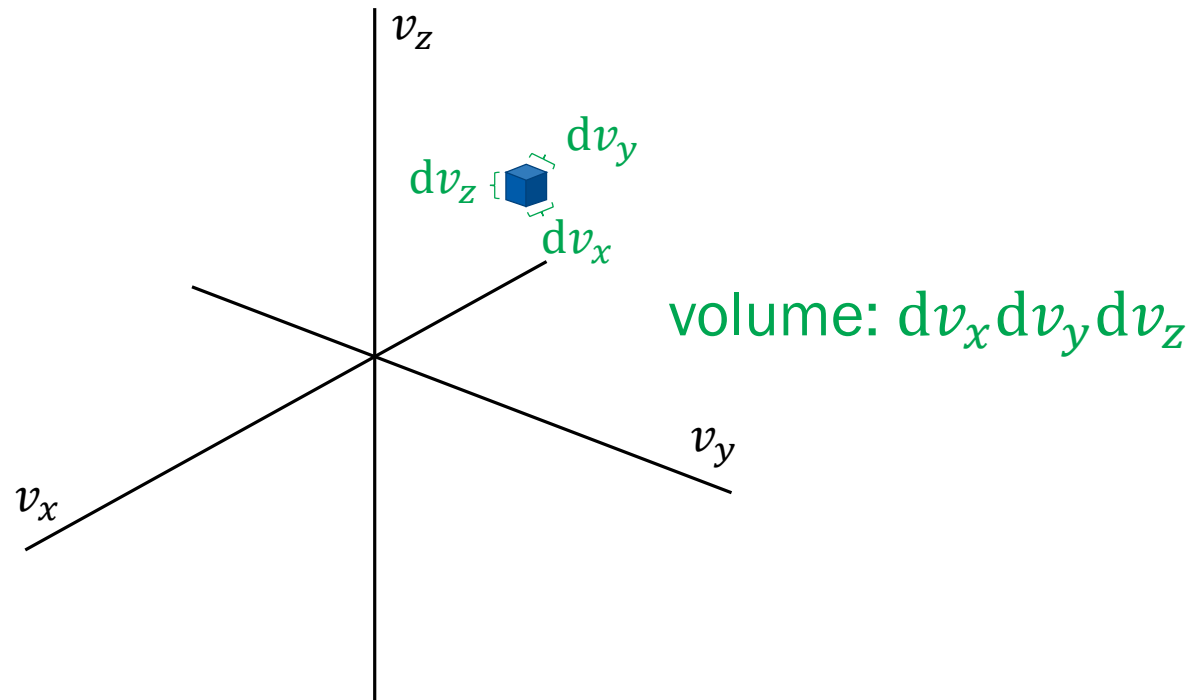
$$f(v_x, v_y, v_z) = \left(\frac{M}{2\pi RT}\right)^{3/2} e^{-M(v_x^2 + v_y^2 + v_z^2)/2RT}$$

- Speed distribution (Maxwell-Boltzmann):

$$f(v) = 4\pi \left(\frac{M}{2\pi RT}\right)^{3/2} v^2 e^{-Mv^2/2RT}$$

Last class

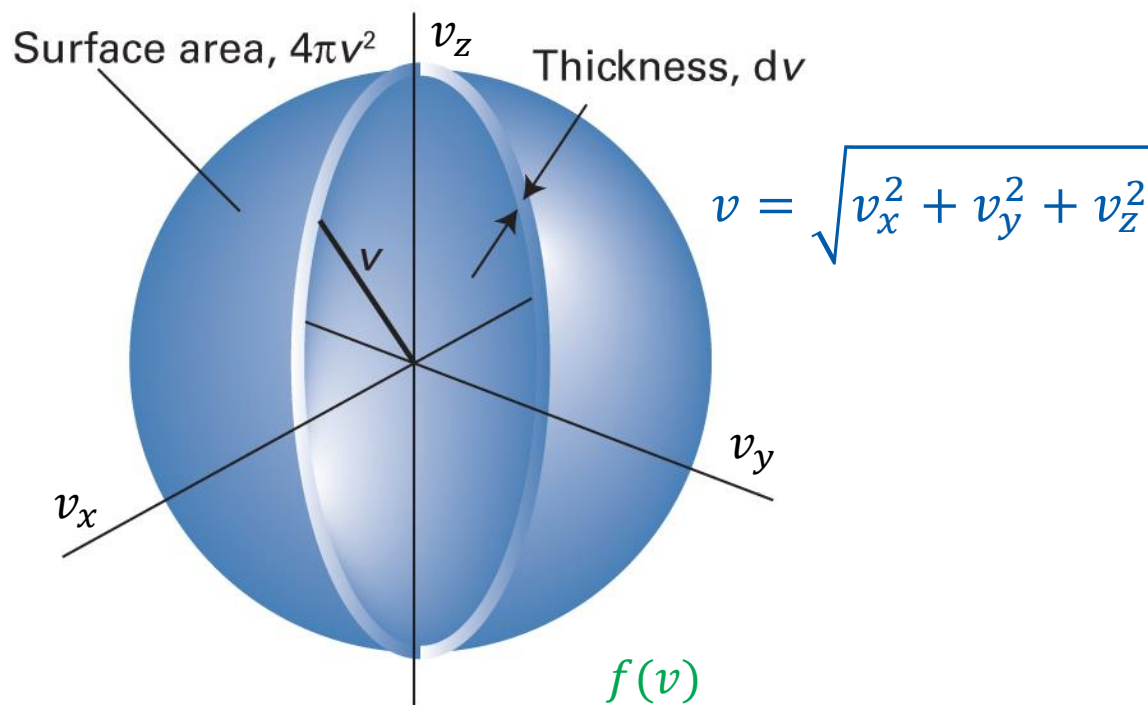
Velocity Space



$$f(v_x, v_y, v_z) dv_x dv_y dv_z = \text{function} \times \text{volume}$$

Last class

Velocity Space



$$\text{function} \times \text{volume} = f(v_x, v_y, v_z) \times 4\pi v^2 dv$$

Last class

Characteristic Speeds: KMT

1. Mean speed

$$\langle v \rangle = \left(\frac{8RT}{\pi M} \right)^{1/2}$$

2. Root-mean-square speed

$$v_{\text{rms}} = \left(\frac{3RT}{M} \right)^{1/2}$$

3. Most probable speed

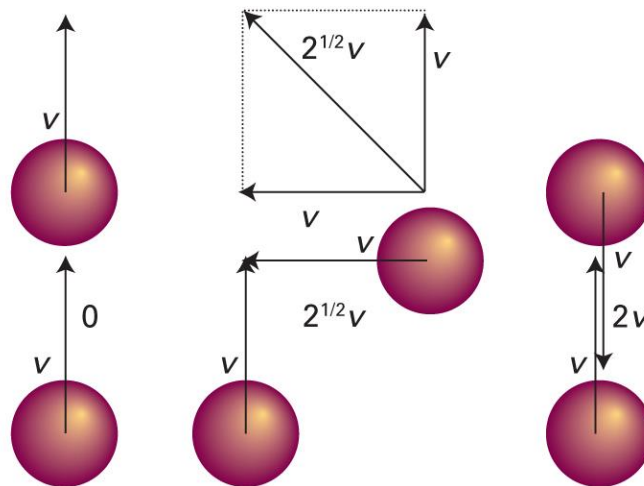
$$v_{\text{mp}} = \left(\frac{2RT}{M} \right)^{1/2}$$



Last class

Mean Relative Speed

“Mean speed with which a molecule approaches another”



$$v_{\text{rel}} = 2^{1/2} \langle v \rangle = \left(\frac{16RT}{\pi M} \right)^{1/2}$$

(Image: Atkins et al., *Atkins' Physical Chemistry*, 12th ed., Figure 1B.7)

Last class

Mean Relative Speed

For two molecules of different types (A and B),

$$v_{\text{rel}} = \left(\frac{8k_{\text{B}}T}{\pi\mu} \right)^{1/2}$$

where reduced mass $\mu = m_{\text{A}}m_{\text{B}}/(m_{\text{A}} + m_{\text{B}})$

Last class

Molecule-Molecule Collisions

- Collision frequency

$$z = \sigma v_{\text{rel}} \frac{n N_A}{V} = \sigma v_{\text{rel}} \frac{p}{k_B T}$$

- Mean free path

$$\lambda = \frac{v_{\text{rel}}}{z} = \frac{k_B T}{\sigma p}$$

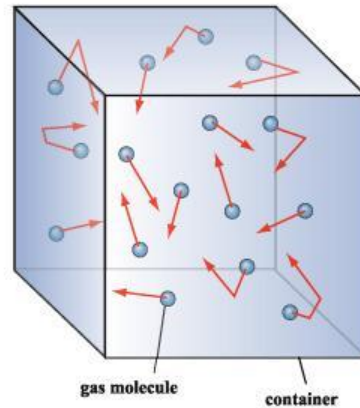
- Collision density for a two-component system

$$Z_{AB} = \sigma v_{\text{rel}} N_A^2 \frac{n_A}{V} \frac{n_B}{V} \propto [A][B]$$



Last class

Pressure



momentum change $\rightarrow$ force $\rightarrow$ pressure

$$pV = \frac{1}{3}nM\langle v^2 \rangle$$

$$pV = nRT$$



(Image: http://web2.uwindsor.ca/courses/physics/high_schools/2005/Brownian_motion/kinetictheory.jpg)

Internal Energy, U

“The total energy of a system”

For a perfect gas,

$$U = \sum_{i=1}^N E_{k,i} = \sum_{i=1}^N \frac{1}{2} m v_i^2 = \frac{1}{2} m \sum_{i=1}^N v_i^2$$

On the other hand,

$$\langle v^2 \rangle = \frac{1}{N_A n} \sum_{i=1}^N v_i^2$$

$$\sum_{i=1}^N v_i^2 = N_A n \langle v^2 \rangle$$

U of Perfect Gas

Substitution leads to

$$U = \frac{1}{2} m N_A n \langle v^2 \rangle = \frac{1}{2} M n \langle v^2 \rangle$$

From the Maxwell-Boltzmann distribution,

$$\langle v^2 \rangle = v_{\text{rms}}^2 = \frac{3RT}{M}$$

Hence,

$$U = \frac{3}{2} nRT$$

Equipartition Theorem

For a sample at thermal equilibrium,
the average value of each quadratic contribution
to the internal energy is $(1/2)k_B T$.

(Quadratic contribution: $E \propto p_x^2$ or $E \propto x^2$)

Monatomic Perfect Gas

- Potential energy = 0
- Kinetic energy: translational energy

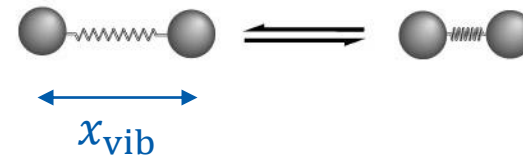
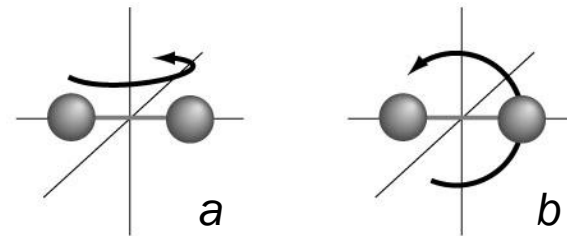
$$E_{\text{k,tr}} = \frac{p^2}{2m} = \frac{p_x^2}{2m} + \frac{p_y^2}{2m} + \frac{p_z^2}{2m}$$

→ 3 quadratic terms per molecule

$$U = (nN_A) \times \left(\frac{3}{2} k_B T \right) = \frac{3}{2} n N_A k_B T = \frac{3}{2} n R T$$

Diatomic Perfect Gas

- Potential energy = 0
- Kinetic energy
 - translation
 - rotation: two possible motions
- vibration: one possible motion



Diatomic Perfect Gas

- Translational energy: 3 quadratic terms per molecule

$$E_{k,\text{tr}} = \frac{p_x^2}{2m} + \frac{p_y^2}{2m} + \frac{p_z^2}{2m}$$

- Rotational energy: 2 quadratic terms per molecule

$$E_{k,\text{rot}} = \frac{p_a^2}{2m} + \frac{p_b^2}{2m}$$

- Vibrational energy: 2 quadratic terms per molecule

$$E_{k,\text{vib}} = \frac{p_{\text{vib}}^2}{2m} + \frac{1}{2} k x_{\text{vib}}^2$$

Diatomic Perfect Gas

- Translational energy: 3 quadratic terms per molecule
- Rotational energy: 2 quadratic terms per molecule
- Vibrational energy: 2 quadratic terms per molecule

Total: 7 quadratic terms per molecule

From the equipartition theorem,

$$U = (nN_A) \times \left(\frac{7}{2} k_B T\right) = \frac{7}{2} nN_A k_B T = \frac{7}{2} nRT$$

Polyatomic Perfect Gas

- Degree of freedom f : number of variables that may vary independently

$$f = 3a \text{ (} a: \text{ number of atoms)}$$

$$\text{CO}_2: f = 3 \times 3 = 9$$

$$\text{H}_2\text{O}: f = 3 \times 3 = 9$$

$$\text{NH}_3: f = 3 \times 4 = 12$$

Polyatomic Perfect Gas

- Linear molecules (e.g. CO₂)
 - Translation: 3 independent motions → 3 quadratic
 - Rotation: 2 independent motions → 2 quadratic
 - Vibration: $(f - 5)$ independent motions → $2(f - 5)$ quadratic

$$U = (2f - 5)nRT/2$$

- Nonlinear molecules (e.g. H₂O)
 - Translation: 3 independent motions → 3 quadratic
 - Rotation: 3 independent motions → 3 quadratic
 - Vibration: $(f - 6)$ independent motions → $2(f - 6)$ quadratic

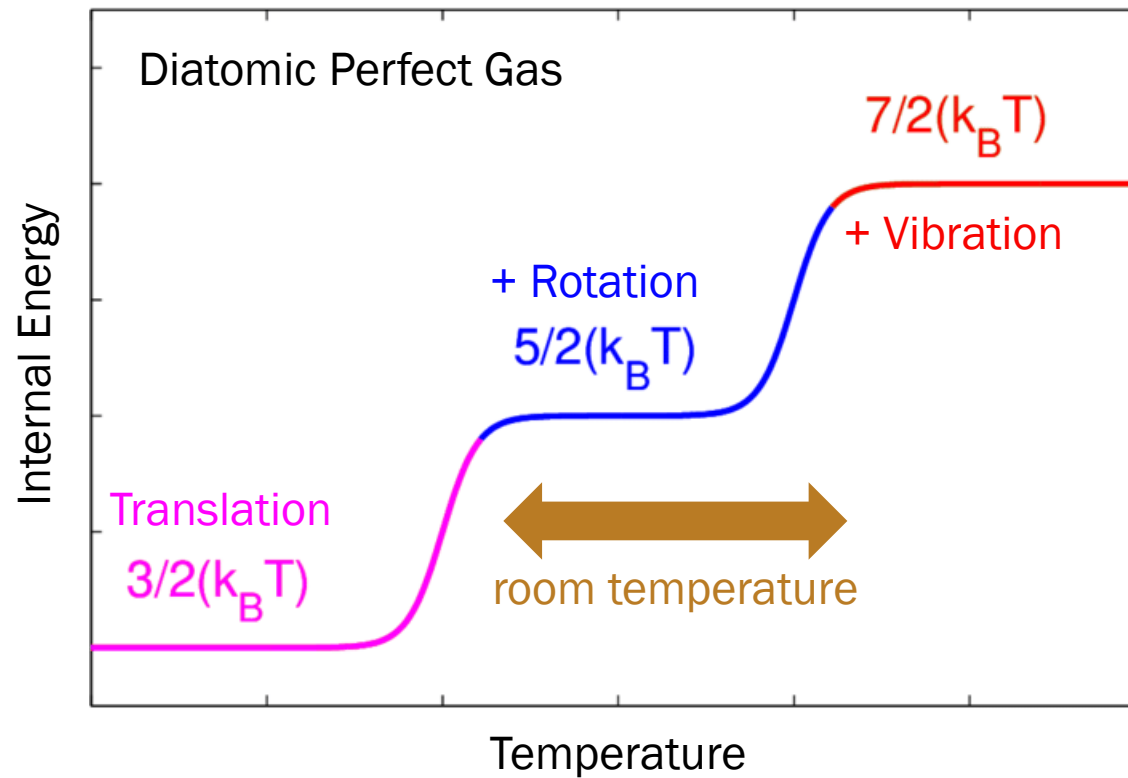
$$U = (2f - 6)nRT/2$$

U of Perfect Gas

The internal energy of a perfect gas is independent of the volume it occupies.

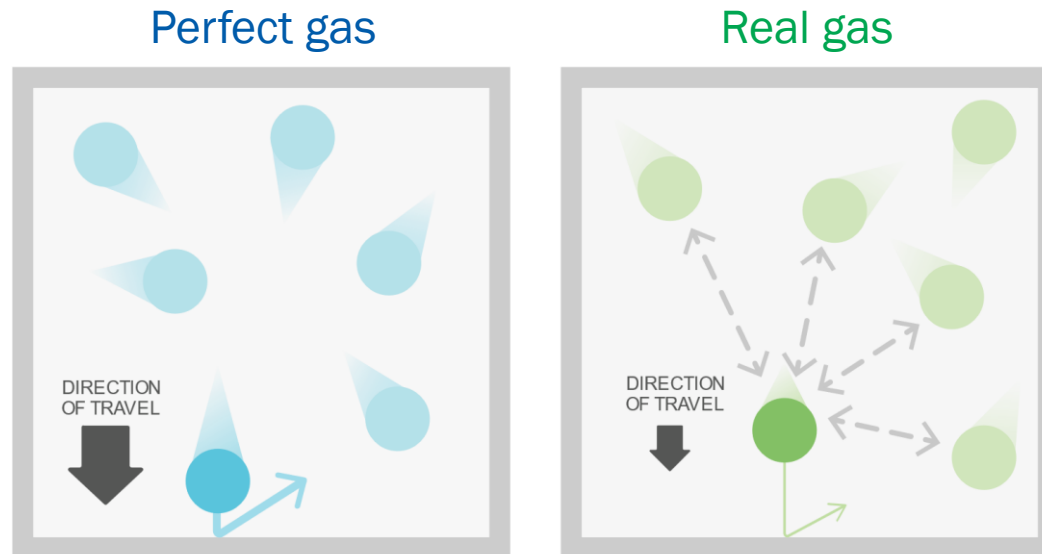
- No molecule-molecule interactions in a perfect gas
- The molecule-molecule distance has no effect on U

Excitation of Motions



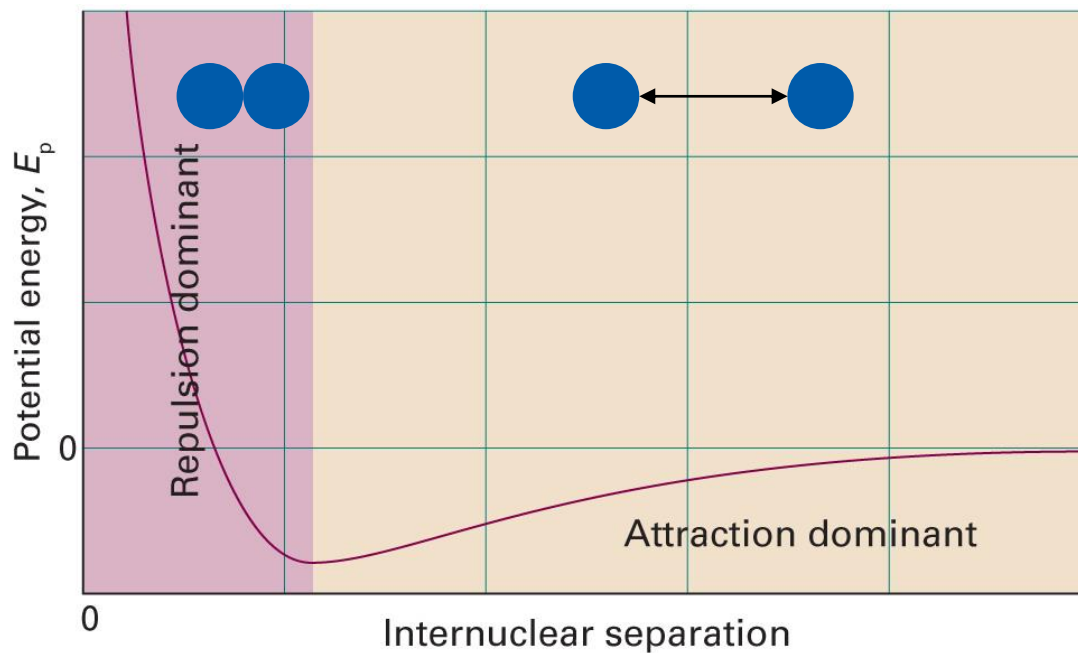
Perfect Gas vs. Real Gas

- Perfect gas (ideal gas): no interactions
- Real gas: finite interactions between molecules



Real Gas

Now the molecules interact!



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 1C.1)

Compression Factor Z

$$Z = V_{\text{m}}/V_{\text{m}}^{\circ}$$

Molar volume of a gas $V_{\text{m}} = V/n$

Molar volume of a perfect gas V_{m}°

Since $V_{\text{m}}^{\circ} = RT/p$,

$$Z = \frac{pV_{\text{m}}}{RT} = \frac{pV}{nRT}$$

Amplitude of Z

$$Z = V_m/V_m^0$$

- $Z > 1$:

$V_m > V_m^0 \rightarrow$ repulsion is dominant

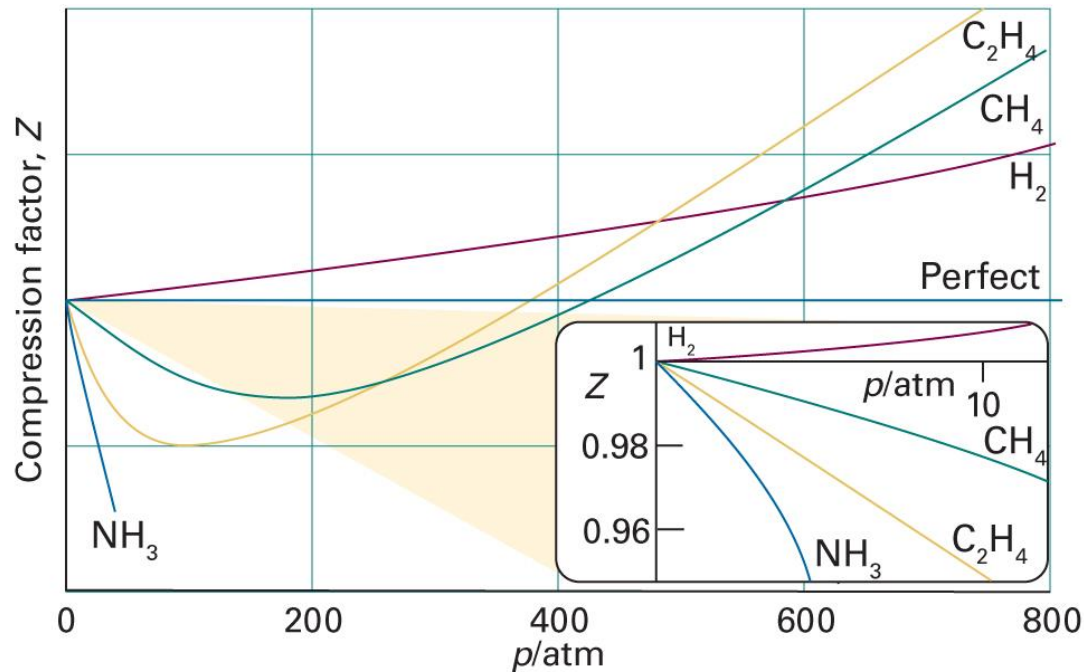
- $Z \approx 1$:

$V_m \approx V_m^0 \rightarrow$ perfect-gas-like behavior

- $Z < 1$:

$V_m < V_m^0 \rightarrow$ attraction is dominant

Experimental Values of Z



$$Z = f(p)?$$

(Image: Atkins et al., *Atkins' Physical Chemistry*, 12th ed., Figure 1C.3)

Virial Expansion

$$Z(p) = 1 + B'p + C'p^2 + \dots$$

$$pV_m = RT(1 + B'p + C'p^2 + \dots)$$

The following expansion is more often used:

$$Z(V_m) = 1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \dots$$

$$pV_m = RT \left(1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \dots \right)$$

Virial Coefficients

$$Z(V_m) = 1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \dots$$

B : second virial coefficient

C : third virial coefficient

$\vdots$

- First virial coefficient = 1
- Virial coefficients are temperature-dependent

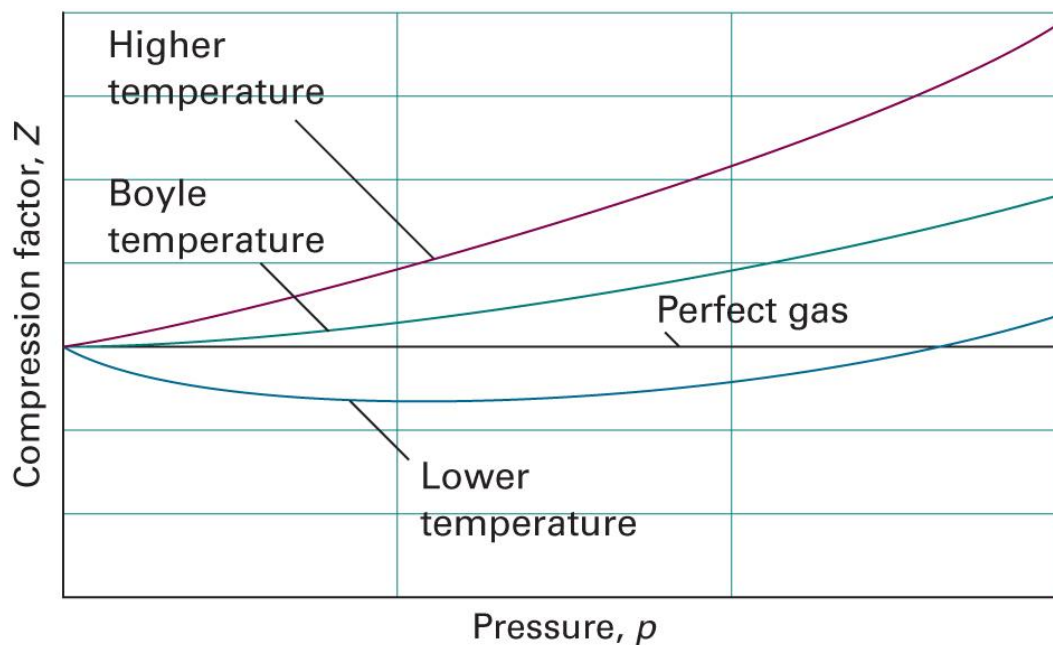
Second Virial Coefficient

$$Z(V_m) = 1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \dots$$

- Generally, $B/V_m \gg C/V_m^2$
- $B > 0 \rightarrow Z > 1 \rightarrow$ repulsion is dominant
- $B = 0 \rightarrow Z \approx 1$ (perfect-gas-like) for a larger region
- $B < 0 \rightarrow Z < 1 \rightarrow$ attraction is dominant

Boyle Temperature T_B

Temperature where $B = 0$



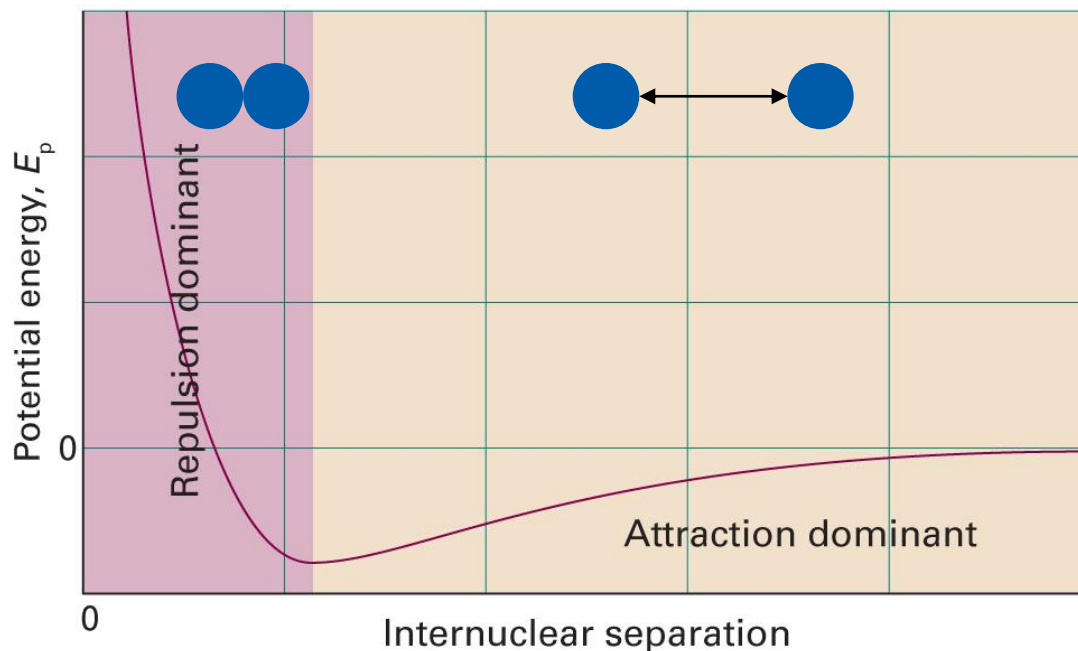
(Image: Atkins et al., *Atkins' Physical Chemistry*, 12th ed., Figure 1C.4)

Experimental Data

	$B / (\text{cm}^3 \text{ mol}^{-1})$		T_B / K
	$T = 273 \text{ K}$	$T = 600 \text{ K}$	
Ar	-21.7	11.9	411.5
CO ₂	-149.7	-12.4	714.8
N ₂	-10.5	21.7	327.2
Xe	-153.7	-19.6	768.0

Corrected Perfect Gas Law

Consider the effects of **repulsion** and **attraction**



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 1C.1)

van der Waals Equation

- **Repulsion:** decrease of the total volume

$$V \rightarrow V - nb$$

$$p = \frac{nRT}{V - nb}$$

- **Attraction:** decrease of the pressure

$\Delta p \sim$ (frequency of collisions) $\times$ (force of each particle)

$$z = \sigma v_{\text{rel}} N_A \frac{n}{V}$$

$$F_{\text{wall}} = \frac{MA\langle v^2 \rangle n}{3V}$$

$$p = \frac{nRT}{V - nb} - a \frac{n^2}{V^2}$$

van der Waals Coefficients

$$p = \frac{nRT}{V - nb} - a \frac{n^2}{V^2}$$

- a : strength of attractive interactions
- b : strength of repulsive interactions
- Characteristics of each gas
- Independent of the temperature

Experimental Data

	$a / (\text{atm L}^2 \text{ mol}^{-2})$	$b / (10^{-2} \text{ L mol}^{-1})$
Ar	1.337	3.20
CO ₂	3.610	4.29
He	0.0341	2.38
Xe	4.137	5.16

Equation Manipulations

$$p = \frac{nRT}{V - nb} - a \frac{n^2}{V^2}$$

$$p = \frac{RT}{V_m - b} - \frac{a}{V_m^2}$$

can be represented by a cubic equation in V_m

van der Waals and Virial

- van der Waals equation

$$p = \frac{RT}{V_m - b} - \frac{a}{V_m^2}$$

- Virial expansion

$$pV_m = RT \left(1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \dots \right)$$

Can we connect the two?

Some Math

Target

$$pV_m = RT \left(1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \dots \right)$$

$$\text{vdW: } p = \frac{RT}{V_m - b} - \frac{a}{V_m^2}$$

$$p = \frac{1}{V_m} \left(\frac{RT}{1 - b/V_m} \right) - \frac{a}{V_m^2}$$

$$pV_m = \frac{RT}{1 - b/V_m} - \frac{a}{V_m}$$

$$pV_m = RT \left(\frac{1}{1 - b/V_m} - \frac{a/RT}{V_m} \right)$$

Some More Math

Target

$$pV_m = RT \left(1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \dots \right)$$

$$pV_m = RT \left(\frac{1}{1 - b/V_m} - \frac{a/RT}{V_m} \right)$$

As $b/V_m \ll 1$, we use

$$\frac{1}{1 - x} = 1 + x + x^2 + \dots \text{ (when } x < 1 \text{)}$$

$$pV_m = RT \left[\left\{ 1 + \frac{b}{V_m} + \left(\frac{b}{V_m} \right)^2 + \dots \right\} - \frac{a/RT}{V_m} \right]$$

Finally

Target

$$pV_m = RT \left(1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \dots \right)$$

$$pV_m = RT \left[\left\{ 1 + \frac{b}{V_m} + \left(\frac{b}{V_m} \right)^2 + \dots \right\} - \frac{a/RT}{V_m} \right]$$

$$pV_m = RT \left[1 + \left(b - \frac{a}{RT} \right) \frac{1}{V_m} + \frac{b^2}{V_m^2} + \dots \right]$$

Hence,

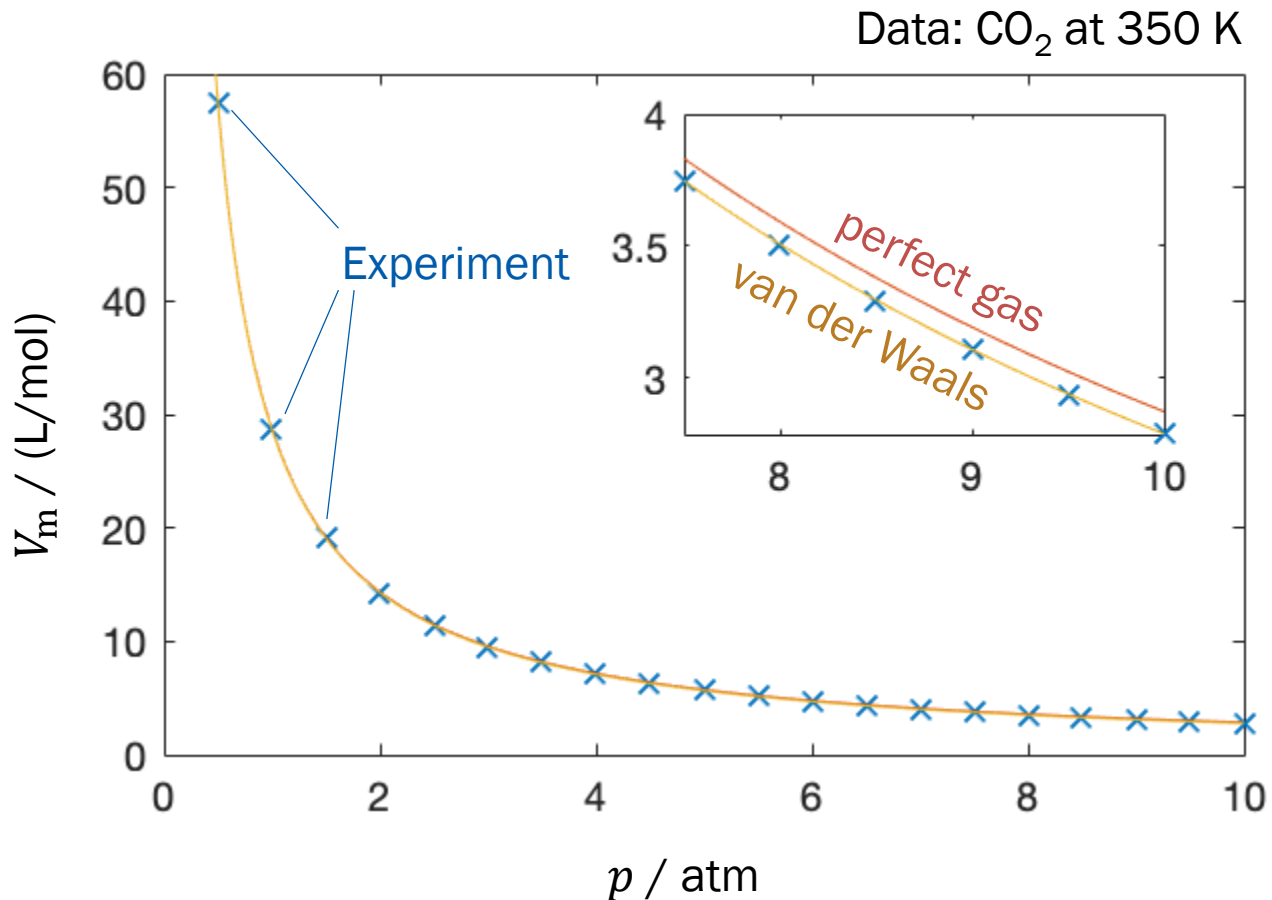
$$B = b - a/RT \text{ and } C = b^2$$



Also, from $B = b - a/RT = 0$, $T_B = a/(bR)$

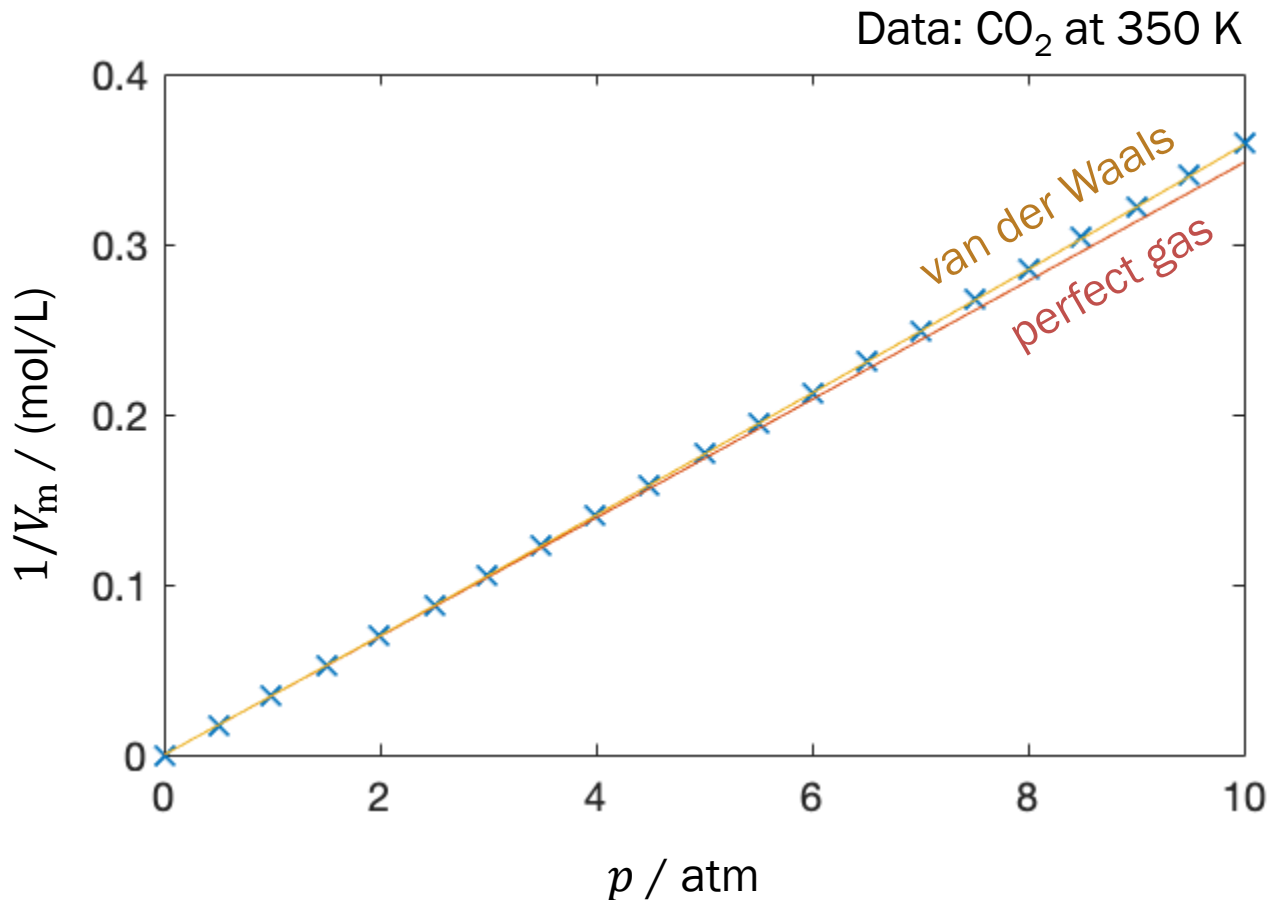
Not in Atkins

Validity of van der Waals



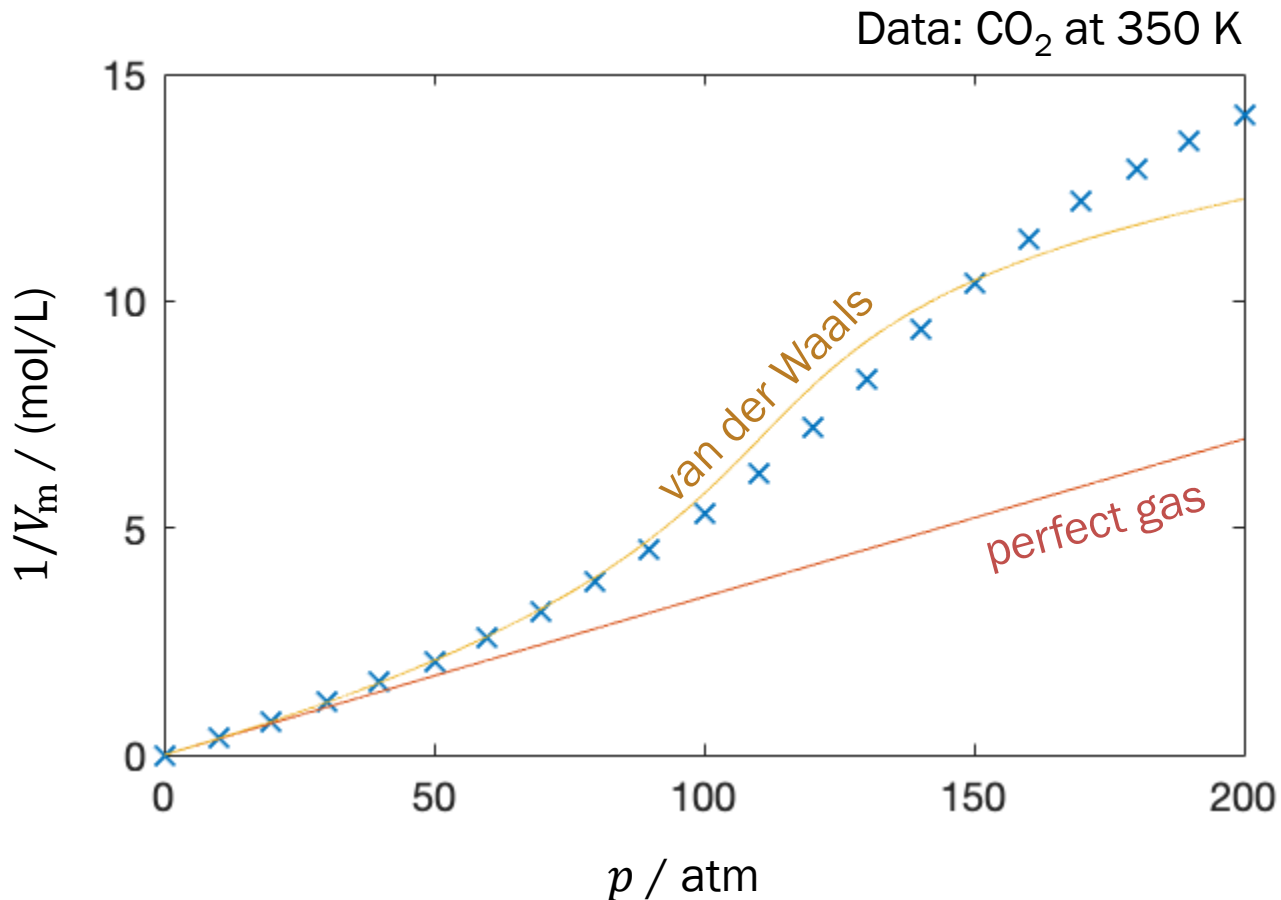
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Validity of van der Waals



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Is van der Waals Perfect?



Other Real Gas Equations

- van der Waals equation (1873)

$$p = \frac{RT}{V_m - b} - \frac{a}{V_m^2}$$

- Modification of the attraction term
 - Redlich-Kwong equation (1949)

$$p = \frac{RT}{V_m - B_{RK}} - \frac{A_{RK}}{V_m(V_m + B_{RK})T^{1/2}}$$

- Peng-Robinson equation (1976)

$$p = \frac{RT}{V_m - B_{PR}} - \frac{A_{PR}}{V_m(V_m + B_{PR}) + B_{PR}(V_m - B_{PR})}$$

Other Real Gas Equations

- van der Waals equation (1873)

$$p = \frac{RT}{V_m - b} - \frac{a}{V_m^2}$$

- Modification of the repulsion term
 - Guggenheim equation (1965)

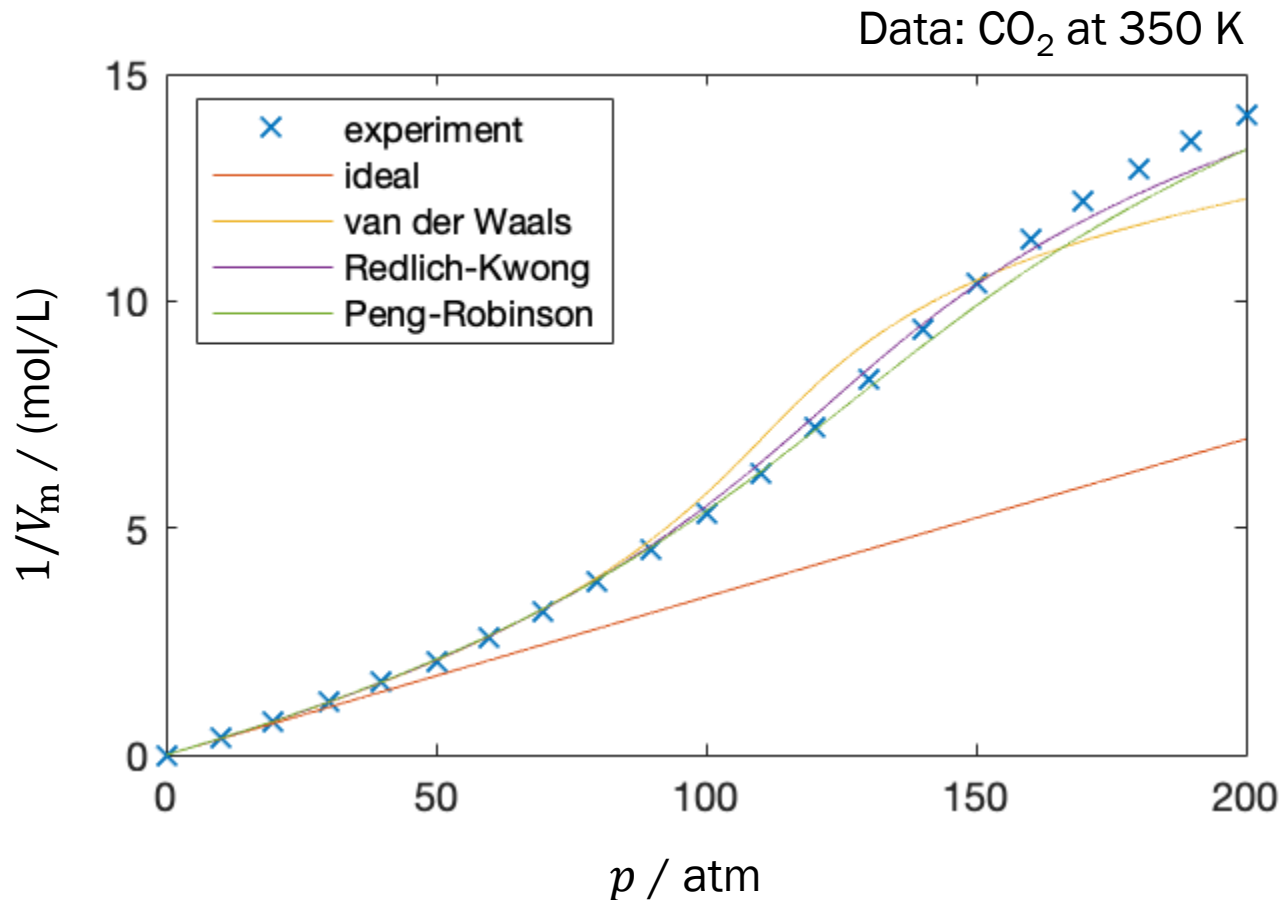
$$p = \frac{RT}{V_m(1 - y)^4} - \frac{A_G}{V_m^2}, \quad y = \frac{B_G}{V_m}$$

- Carnahan-Starling-van der Waals equation (1972)

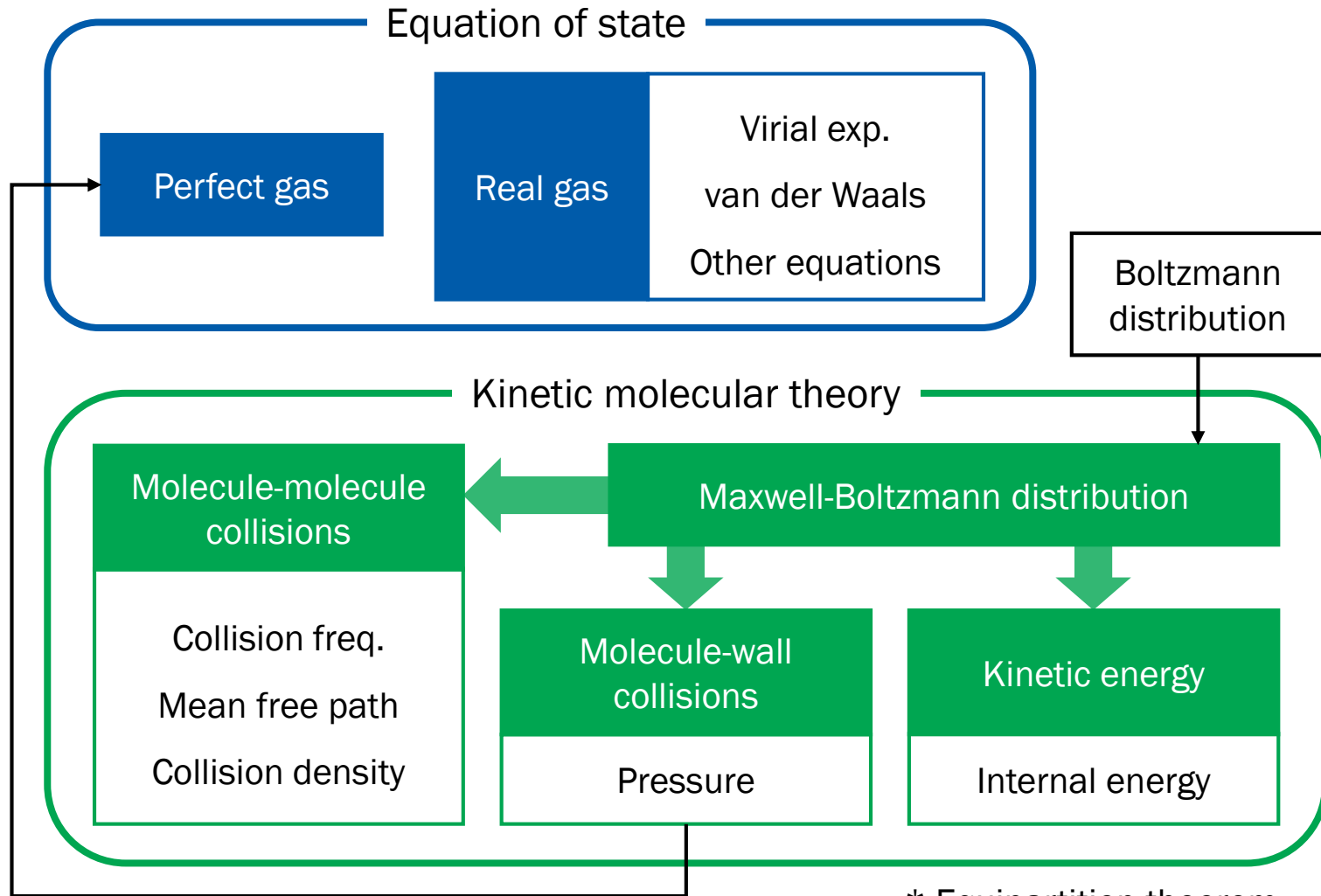
$$p = \frac{RT(1 + y + y^2 - y^3)}{V_m(1 - y)^3} - \frac{A_{CSW}}{V_m^2}, \quad y = \frac{B_{CSW}}{V_m}$$

Not in Atkins

Better Descriptions



Summary



* Equipartition theorem

Where Are We?

